Outcomes and content for Stage 4

Computation with integers

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- compares, orders and calculates with integers to solve problems **MA4-INT-C-01**

Related Life Skills outcomes: MALS-LAN-01, MALS-LAN-02, MALS-COU-01, MALS-REP-01, MALS-COM-01, MALS-ADS-01, MALS-MDI-01

Content

Compare and order integers

- Recognise and describe the direction and magnitude of integers
- Identify and represent integers on a number line
- Compare the relative value of integers using the less than (<) and greater than (>) symbols
- Order integers

Add and subtract positive and negative integers

- Add and subtract integers with and without the use of digital tools
- Construct a directed number sentence to model a situation
- Examine different meanings (position or operation) for the $+$ and $-$ signs, depending on context

Multiply and divide positive and negative integers

- Represent multiples of negative integers as repeated addition
- Multiply and divide positive and negative integers with and without the use of digital tools

Apply the 4 operations to integers

- Apply the 4 operations to integers
- Solve problems involving grouping symbols with integers
- Apply the order of operations to evaluate expressions involving integers, with and without the use of digital tools

Fractions, decimals and percentages

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- represents and operates with fractions, decimals and percentages to solve problems **MA4-FRC-C-01**

Related Life Skills outcomes: MALS-FRC-01, MALS-DEP-01

Content

Compare fractions using equivalence

- Determine the highest common factor (HCF) of 2 whole numbers
- Examine methods of generating equivalent fractions
- Simplify fractions by using methods, including determining the HCF of the numerator and denominator or repeated simplification using common factors
- Create fractions with the same denominator to compare their sizes
- Compare and order fractions with different denominators

Round decimals to a specified degree of accuracy using approximations

- Round decimals to a given number of decimal places
- Apply the notation ≈ as a symbol of numerical approximation
- Reason why an approximation may be more appropriate than an exact answer and vice versa

Identify terminating and recurring decimals

- Use either dot or vinculum notation for recurring (repeating) decimals
- Classify decimals as recurring or terminating

Identify and make use of the relationship between fractions, decimals and percentages to carry out simple conversions

- Define rational numbers as numbers that can be written in the form $\frac{a}{b}$, where a and b are integers and $b \neq 0$
- Classify fractions and percentages as rational numbers
- Recognise and explain that numbers with terminating or recurring decimals are rational
- Represent fractions as decimals (terminating and recurring) and percentages
- Represent terminating decimals as fractions and percentages
- Represent improper fractions as mixed numbers and decimals, and vice versa
- Represent percentages as fractions and decimals

Examine the concept of irrational numbers

- Identify and define irrational numbers as numbers that cannot be written in the form $\frac{a}{b}$ where a and b are integers and $b \neq 0$
- Find approximations of irrational numbers using digital tools
- Locate the approximate position of irrational numbers on a number line

Order and compare the value of fractions, decimals and percentages

- Locate positive and negative fractions, decimals and mixed numbers on a number line to compare their relative values
- Compare and order fractions, mixed numbers, decimals (terminating and recurring) and percentages

Solve problems that involve the addition and subtraction of fractions

- Represent addition and subtraction of fractions with the same or unrelated denominators
- Solve problems involving adding and subtracting fractions and mixed numbers, including finding a common denominator
- Solve problems that involve subtracting a fraction from a whole number, with and without the use of digital tools

Solve problems that involve the multiplication and division of fractions and decimals

- Compare and generalise the effect of multiplying or dividing by a number with magnitude between zero and one
- Represent multiplication and division of decimals
- Represent multiplication and division of fractions, including mixed numbers
- Multiply and divide decimals, using digital tools to solve problems
- Multiply and divide fractions and mixed numbers, with and without using digital tools to solve problems
- Compare initial estimates with the results of calculations
- Apply knowledge of fractions and decimals of quantities to solve problems
- Apply knowledge of multiplication and division of fractions and decimals to solve problems

Represent one quantity as a fraction, decimal or percentage of another, with and without the use of digital tools

- Represent one quantity as a fraction, decimal or percentage of another by considering appropriate units
- Calculate percentage increases and decreases in various contexts
- Examine the financial applications of percentage increase and decrease, including profit and/or loss as a percentage of cost price or selling price

Solve problems that involve the use of percentages

- Apply knowledge of percentages to calculate quantities in various contexts
- Apply knowledge of percentage increases and decreases to solve problems in various contexts
- Solve real-life problems involving percentages using the unitary method or other techniques
- Solve financial problems involving percentages, specifically considering GST, profit and loss

Ratios and rates

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving ratios and rates, and analyses distance–time graphs **MA4-RAT-C-01**

Related Life Skills outcomes: MALS-ADS-01, MALS-MDI-01

Content

Recognise and simplify ratios

- Use ratios to compare 2 or more quantities measured in the same units
- Identify and express one part of a ratio as a fraction of the whole
- Simplify ratios

Solve problems involving ratios

- Apply the unitary method to solve ratio problems
- Divide a quantity in a given ratio
- Solve real-life problems involving ratios

Recognise and simplify rates

- Explain the differences between ratios and rates
- Represent given information as a simplified rate
- Convert between units for rates

Solve problems involving rates

- Solve a variety of real-life problems involving rates
- Examine financial applications of rates, such as best buys

Interpret and construct distance–time graphs from authentic data

- Interpret distance–time graphs made up of straight-line segments with a negative, zero or positive slope
- Calculate speeds for straight-line segments of given distance–time graphs
- Create distance–time graphs made up of straight-line segments

Algebraic techniques

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- generalises number properties to operate with algebraic expressions including expansion and factorisation **MA4-ALG-C-01**

Related Life Skills outcomes: MALS-PAT-01

Content

Examine the concept of pronumerals as a way of representing numbers

- Examine and recognise that pronumerals can be used to represent one or more numerical values and when pronumerals have more than one numerical value, they may then be referred to as variables
- Identify and define an algebraic expression as an expression formed by combining numbers and algebraic symbols using arithmetic operations
- Use concise algebraic notation and conventions for multiplication, division and powers, and explain the meanings for each convention

Create algebraic expressions and evaluate them by substitution

- Generate algebraic expressions by translating descriptions and vice versa
- Substitute numbers into algebraic expressions and evaluate the result
- Generate a number pattern from an algebraic expression

Extend and apply the laws and properties of arithmetic to algebraic terms and expressions

- Generalise the associative property of addition and multiplication to algebraic expressions
- Generalise the commutative property to algebraic expressions
- Identify like terms, and add and subtract them to simplify algebraic expressions
- Simplify algebraic expressions that involve multiplication and division, including simple algebraic fractions
- Simplify algebraic expressions involving mixed operations

Extend and apply the distributive law to the expansion of algebraic expressions

- Explain the role and meaning of grouping symbols in algebraic expressions
- Apply the distributive law to expand and simplify algebraic expressions by removing grouping symbols

Factorise algebraic expressions by identifying numerical and algebraic factors

- Identify and list factors of a single term
- Factorise algebraic expressions using knowledge of factors and finding the highest common numerical factor (HCF)
- Factorise algebraic expressions using knowledge of factors by finding a common algebraic factor, including expressions involving more than 2 terms, and verify the result by expansion

Indices

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- operates with primes and roots, positive-integer and zero indices involving numerical bases and establishes the relevant index laws **MA4-IND-C-01**

Content

Apply index notation to represent whole numbers as products of powers of prime numbers

- Describe numbers written in index form using terms such as *base*, *power*, *index* and *exponent*
- Represent numbers in index notation limited to positive powers
- Represent in expanded form and evaluate numbers expressed in index notation, including powers of 10
- Apply the order of operations to evaluate expressions involving indices
- Determine and apply tests for divisibility for 2, 3, 4, 5, 6 and 10
- Represent a whole number greater than one as a product of its prime factors, using index notation where appropriate

Examine cube roots and square roots

- Use the notations for square root ($\sqrt{\ }$) and cube root ($\sqrt[3]{\ }$)
- Recognise and describe the relationship between squares and square roots, and cubes and cube roots for positive numbers
- Verify, through numerical examples, that $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$
- Estimate the square root of any non-square whole number and the cube root of any non-cube whole number, then check using a calculator
- Identify and describe exact and approximate solutions in the context of square roots and cube roots
- Apply the order of operations to evaluate expressions involving square roots, cube roots, square numbers and cube numbers

Use index notation to establish the index laws with positive-integer indices and the zero index

- Establish the multiplication, division and the power of a power index laws, by expressing each number in expanded form with numerical bases and positive-integer indices
- Verify through numerical examples that $(ab)^2 = a^2b^2$
- Establish the meaning of the zero index
- Apply index laws to simplify and evaluate expressions with numerical bases

Equations

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves linear equations of up to 2 steps and quadratic equations of the form $ax^2 = c$ **MA4-EQU-C-01**

Related Life Skills outcomes: MALS-ADS-01, MALS-MDI-01

Content

Solve linear equations up to 2 steps

- Represent number sentences involving unknown quantities using pronumerals
- Describe number sentences as equations
- Distinguish between and compare algebraic expressions and equations
- Solve linear equations with integer and non-integer solutions using algebraic techniques that involve up to 2 steps, including equations with pronumerals on both sides
- Model and solve word problems using equations of up to 2 steps

Solve and verify linear equations by substitution

- Verify solutions to equations by substitution
- Solve problems involving linear equations, including those arising from substituting given values into formulas

Solve quadratic equations

- Reason why there are 2 values of x that satisfy a quadratic equation of the form $x^2 = c$ if $c > 0$
- Solve problems involving quadratic equations of the form $ax^2 = c$, giving answers in exact form and as decimal approximations
- Solve quadratic equations arising from substitution into a formula

Linear relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- creates and displays number patterns and finds graphical solutions to problems involving linear relationships **MA4-LIN-C-01**

Related Life Skills outcomes: MALS-POS-01

Content

Plot and identify points on the Cartesian plane

- Plot and label points on the Cartesian plane of given coordinates, including those with coordinates that are not whole numbers
- Identify and record the coordinates of given points on the Cartesian plane, including those with coordinates that are not whole numbers

Plot linear relationships on the Cartesian plane

- Construct a geometric pattern and record the results in a table of values
- Represent a given number pattern (including decreasing patterns) using a table of values
- Describe a number pattern in words and generate an equation using algebraic symbols
- Apply an equation generated from a pattern to calculate the corresponding value for a smaller or larger number
- Recognise that a linear relationship can be represented by a number pattern, an equation (or a rule using algebraic symbols), a table of values, a set of pairs of coordinates and a line graphed on a Cartesian plane, and move flexibly between these representations
- Explain that there are an infinite number of ordered pairs that satisfy a given linear relationship by extending a line joining a set of points on the Cartesian plane
- Compare similarities and differences of multiple straight-line graphs on the same set of axes using graphing applications
- Describe linear relationships in real-life contexts and solve related problems

Solve linear equations using graphical techniques

- Recognise that each point on the graph of a linear relationship satisfies the equation of a line
- Apply graphs of linear relationships to solve a corresponding linear equation using graphing applications
- Graph 2 intersecting lines on the same set of axes and identify the point of intersection using either graphing applications or a table of values
- Verify that the point of intersection satisfies the equations of both lines

Length

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of the perimeter of plane shapes and the circumference of circles to solve problems **MA4-LEN-C-01**

Related Life Skills outcomes: MALS-LEN-01

Content

Solve problems involving the perimeter of various quadrilaterals and simple composite figures

- Solve problems involving the perimeter of plane shapes, including parallelograms, trapeziums, rhombuses and kites
- Solve problems relating to the perimeter of simple composite figures
- Compare methods of solution for finding perimeter and evaluate the efficiency of those methods

Describe the relationships between the features of circles

- Identify and describe the relationship between circle features, including the radius, diameter, arc, chord, sector and segment of a circle, and a tangent to a circle
- Define π as the ratio of the circumference to the diameter of any circle
- Verify that the number π is a constant and develop the formula for the circumference of a circle
- Apply the formula for the circumference of a circle in terms of the diameter d or radius r (circumference of a circle $= \pi d$ or $2\pi r$) to solve related problems to solve related problems
- Establish the arc length formula $\left(l = \frac{\theta}{360} \times 2\pi r\right)$ where l is the arc length and θ is the angle subtended at the centre by the arc
- Solve problems by finding arc lengths and the perimeter of sectors, giving an exact answer in terms of π or an approximate answer
- Find the perimeter of quadrants, semicircles and simple composite figures consisting of 2 shapes in a variety of contexts, including using digital tools

Right-angled triangles (Pythagoras’ theorem)

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies Pythagoras’ theorem to solve problems in various contexts **MA4-PYT-C-01**

Content

Identify and define Pythagoras’ theorem

- Identify and describe the hypotenuse as the side opposite the right angle and the longest side in any right-angled triangle
- Establish the relationship between the lengths of the sides of a right-angled triangle
- Use the relationship to record and define Pythagoras’ theorem both algebraically and in words

Examine problems involving Pythagoras’ theorem

- Apply Pythagoras’ theorem to find the unknown length of a side in a right-angled triangle, giving answers in an exact form or as decimal approximations
- Apply the converse of Pythagoras’ theorem to establish whether a triangle is right angled
- Solve practical problems involving Pythagoras’ theorem before exploring a variety of related problems
- Justify whether a set of 3 integers is a Pythagorean triad

Area

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of area and composite area involving triangles, quadrilaterals and circles to solve problems **MA4-ARE-C-01**

Related Life Skills outcomes: MALS-ARE-01

Content

Develop and use formulas to find the area of rectangles, triangles and parallelograms to solve problems

- Apply the formula to find the area of a rectangle or square: $A = lb$, where l is the length and b is the breadth (or width) of the rectangle or square
- Develop and apply the formula to find the area of a triangle: $A = \frac{1}{2}bh$, where b is the base length and h is the perpendicular height
- Develop and apply the formula to find the area of a parallelogram: $A = bh$ where b is the base length and h is the perpendicular height
- Calculate the area of composite figures that can be dissected into rectangles, squares, parallelograms or triangles to solve problems

Develop and use the formula to find the area of circles and sectors to solve problems

- Develop and apply the formula to find the area of a circle: $A = \pi r^2$, where r is the length of the radius
- Explain how the area of a sector can be developed from the area of a circle $\left(A = \frac{\theta}{360} \times \pi r^2\right)$
- Find the area of quadrants, semicircles and sectors, and apply these formulas in the context of real-life problems
- Calculate the areas of composite shapes involving quadrants, semicircles and sectors to solve problems

Develop and use the formulas to find the area of trapeziums, rhombuses and kites to solve problems

- Develop and apply the formula to find the area of a kite or rhombus: $A = \frac{1}{2}xy$, where x and y are the lengths of the diagonals
- Develop and apply the formula to find the area of a trapezium: $A = \frac{h}{2}(a + b)$, where h is the perpendicular height and a and b are the lengths of parallel sides
- Calculate the area of composite shapes involving trapeziums, kites and rhombuses to solve problems

Choose appropriate units of measurement for area and convert between units

- Choose an appropriate unit to measure the area of different shapes and surfaces, and justify the choice

- Convert between metric units of area using $1\ \text{cm}^2 = 100\ \text{mm}^2$, $1\ \text{m}^2 = 10\,000\ \text{cm}^2$, $1\ \text{ha} = 10\,000\ \text{m}^2$ and $1\ \text{km}^2 = 1\,000\,000\ \text{m}^2 = 100\ \text{ha}$

Volume

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies knowledge of volume and capacity to solve problems involving right prisms and cylinders **MA4-VOL-C-01**

Related Life Skills outcomes: MALS-VOL-01

Content

Describe the different views of prisms and solids that have been formed from prism combinations

- Represent prisms from different views in 2 dimensions, including top, side, front and back views
- Describe and illustrate solids formed from prism combinations from different views in 2 dimensions, including top, side, front and back views
- Identify and illustrate the cross-sections of different prisms
- Examine the idea that prisms have a uniform cross-section that is equal to the base area
- Determine if a particular solid has a uniform cross-section

Develop and apply the formula to find the volume of a prism to solve problems

- Develop the formula for the volume of a prism: $V = \textit{base area} \times \textit{height}$, leading to $V = Ah$
- Apply the formula for the volume of a prism to prisms with uniform cross-sections to solve problems

Develop the formula for finding the volume of a cylinder and apply the formula to solve problems

- Develop and apply the formula to solve problems involving the volume of cylinders: $V = \pi r^2 h$, where r is the length of the radius of the base and h is the perpendicular height

Choose appropriate units of measurement for volume and capacity and convert between units

- Recognise that 1000 L is equal to 1 kilolitre (kL) and use the abbreviation
- Recognise that 1000 kL is equal to 1 megalitre (ML) and use the abbreviation
- Choose an appropriate unit to measure the volume or capacity of different objects and justify the choice
- Convert between metric units of volume and capacity (1 cm^3 = 1000 mm^3, 1 cm^3 = 1 mL, 1 m^3 = 1000 L = 1 kL, 1000 kL = 1 ML)
- Solve practical problems involving the volume and capacity of right prisms and cylinders

Angle relationships

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- applies angle relationships to solve problems, including those related to transversals on sets of parallel lines **MA4-ANG-C-01**

Content

Apply the language, notation and conventions of geometry

- Use appropriate terminology and conventions to define, label and name points, rays, lines and intervals using capital letters
- Identify and label the vertex and arms of an angle with capital letters
- Use appropriate conventions to label and name angles
- Use common conventions to indicate right angles, equal angles and intervals on diagrams

Identify geometrical properties of angles at a point

- Identify right angles, straight angles, angles of complete revolution and vertically opposite angles
- Apply the terms *complementary* and *supplementary* to a pair of angles adding to $90°$ and $180°$, respectively
- Apply the term *adjacent angles* to a pair of angles with a common arm and common vertex

Identify and describe corresponding, alternate and co-interior angles when 2 straight lines are crossed by a transversal, including parallel lines

- Identify and describe perpendicular lines using the symbol for *is perpendicular to* ($\perp$)
- Apply the common conventions to indicate parallel lines on diagrams
- Identify and describe pairs of parallel lines using the symbol for *is parallel to* ($\parallel$)
- Identify and define transversals, including transversals of parallel lines
- Identify, name and measure alternate angle pairs, corresponding angle pairs and co-interior angle pairs for 2 lines cut by a transversal
- Verify and identify corresponding angles and alternate angles as equal, and co-interior angles as supplementary, when a pair of parallel lines is cut by a transversal
- Justify that 2 lines are parallel by using properties of alternate, corresponding or co-interior angles on parallel lines

Solve numerical problems involving angles using reasoning

- Apply the knowledge of angle relationships including angles at a point to find the sizes of unknown angles embedded in diagrams and give reasons
- Apply the knowledge of angles associated with parallel lines to find the sizes of unknown angles embedded in related diagrams and give reasons

Properties of geometrical figures

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- identifies and applies the properties of triangles and quadrilaterals to solve problems **MA4-GEO-C-01**

Related Life Skills outcomes: MALS-GEO-01

Content

Classify triangles according to their side and angle properties

- Label triangles using appropriate text and symbols
- Classify and describe types of triangles based on their properties, including acute-angled, right-angled, obtuse-angled, equilateral, isosceles and scalene triangles

Classify quadrilaterals and describe their properties

- Identify quadrilaterals using naming conventions
- Distinguish between convex and non-convex quadrilaterals
- Verify and describe the properties of the special quadrilaterals which include parallelograms, rectangles, rhombuses, squares, trapeziums and kites
- Identify and label the properties of the special quadrilaterals using appropriate conventions
- Classify quadrilaterals based on their properties
- Justify why some quadrilaterals may be classified as more than one type of quadrilateral

Apply the properties of triangles and quadrilaterals

- Prove that the interior angle sum of a triangle is 180° with or without digital tools
- Prove that any exterior angle of a triangle equals the sum of the 2 interior opposite angles
- Apply the angle sum of a triangle to prove that the angle sum of a quadrilateral is 360°
- Apply the properties of triangles and quadrilaterals to determine unknown sides and angles to solve problems, giving reasons

Data classification and visualisation

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- classifies and displays data using a variety of graphical representations **MA4-DAT-C-01**

Related Life Skills outcomes: MALS-DAT-01

Content

Classify data as either numerical (discrete or continuous) or categorical (nominal or ordinal) variables

- Define a variable in the context of statistics as any characteristic, number or quantity that can be measured or counted
- Classify and describe variables as numerical or categorical
- Describe a numerical variable as either discrete or continuous
- Describe a categorical variable as nominal or ordinal
- Distinguish between and compare numerical (discrete or continuous) and categorical (nominal or ordinal) variables

Display data using graphical representations relevant to the purpose of the data

- Represent single datasets using graphs, including frequency histograms and polygons, dot plots, stem-and-leaf plots, divided bar graphs, column graphs, line graphs, sector graphs and pictograms, with or without digital tools
- Include sources, titles, labels and scales when displaying data in a graph
- Select the type of graph best suited to represent various single datasets and justify the choice of graph
- Represent a dataset using a statistical infographic and justify the choice of graphical representation used

Interpret data in graphical representations

- Identify and interpret data displayed on graphs
- Identify features of graphical representations to draw conclusions
- Interpret patterns in graphical representations to make predictions
- Explain why a given graphical representation can lead to a misinterpretation of data

Data analysis

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- analyses simple datasets using measures of centre, range and shape of the data **MA4-DAT-C-02**

Related Life Skills outcomes: MALS-DAT-02

Content

Calculate and compare the mean, median, mode and range for simple datasets

- Calculate the mean $(\bar{x})$ of a set of data using digital tools
- Calculate and describe the mean, median, mode and range of a dataset
- Classify the mean, median and/or mode as measure(s) of centre to represent the average or typical value of a dataset
- Describe and interpret data displays using mean, median and range
- Identify and describe datasets as having no modes (uniform), one mode (unimodal), 2 modes (bimodal) or multiple modes (multimodal)
- Identify the range as a measure of spread to describe variation in a dataset
- Compare simple datasets using the mean, median, mode and range

Interpret the effect individual data points have on measures of centre and range

- Informally identify clusters, gaps and outliers in datasets and give reasons for their occurrence in the context of the data
- Identify and explain the impact of adding or removing data values that are clustered at one end of a dataset on the measures of centre
- Identify and explain the impact of outliers on the measures of centre and range
- Determine and justify the most appropriate measure of centre to summarise the data in its context

Analyse datasets presented in various ways and draw conclusions

- Identify and describe the shape and distribution of a dataset using the terms *symmetrical*, *negatively skewed* and *positively skewed*
- Define a census as a study of every unit, everyone or everything in a population
- Define a sample as a subset of units in a population selected to represent all units in a population of interest
- Draw conclusions and make informed decisions about data gathered using data-collection techniques, including census and sampling, which is then presented in tables, graphs and charts

Probability

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- solves problems involving the probabilities of simple chance experiments **MA4-PRO-C-01**

Related Life Skills outcomes: MALS-PRO-01

Content

Determine probabilities for chance experiments

- List the sample space for chance experiments
- Express the probability of an event, which has a finite number of equally likely outcomes, as $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$
- Recognise that probabilities range from 0 (impossible) to 1 (certain) and that equally likely outcomes have equal probabilities
- Verify that the total of the probabilities of all possible outcomes of an event is 1
- Identify and describe theoretical (expected) probabilities as being the likelihood of outcomes occurring under fair or unbiased conditions
- Explain that observed probability is the relative frequency resulting from repeated trials of a simulation and determine observed probabilities
- Explore relative frequencies by using a random number generator to repeat a chance experiment a number of times

Determine probabilities for complementary events

- Identify and describe the complement of an event
- Verify that the sum of the probability of an event and its complement is a total of 1
- Solve problems involving the probability of complementary events
- Represent the possible outcomes for complementary events in various forms